ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 15

11 July 2025

Overview:

In today's session, we explored the Llama model, a powerful family of large language models developed by Meta. These models are designed to perform various natural language processing tasks like text generation, translation, summarization, and question answering — all with human-like fluency.

Learning Objectives:

- Understand what Llama is and its purpose
- Learn how Llama differs from other large language models like GPT
- Know where Llama is used in real-world AI applications
- Get an idea of how Llama models are trained and scaled

What is Llama?

A line of advanced language models Known as Llama (Large Language Model Meta AI) by Meta aims to grasp and create human-like texts, which makes them valuable tools in several ways such as natural language processing, text generation, and conversational AI.

Key Features of Llama Models:

- Open weight
- Performs well even on smaller GPUs
- Focuses on transparency and responsible AI use
- Scalable from LLaMA-7B to LLaMA-65B parameters

Installation Steps (using Llama as an example):

1. 1. Download and Install Ollama: Begin by downloading the Ollama installer from the official website (e.g., ollama.com) and follow the installation wizard for your operating system.
2. 2. Initialize the Ollama Server: Open your terminal or command prompt and run the command ollama serve to start the Ollama inference server.
3. 3. Download the Llama Model: Use the ollama run command followed by the model's name (e.g., ollama run llama2 or ollama run llama3: latest) to download and start the model.
4. 4. Interact with the Model: We can then interact with the model through the command line or by using tools like CodeGPT in VSCode, which can be configured to use Ollama as a model provider.

Why Llama Matters:

It allows researchers and developers to build high-quality NLP applications while keeping resource costs low. It supports the trend of open-source AI, helping more people understand and improve language models.

Applications of Llama

- Text generation
- Conversational AI
- Machine Translation
- Sentiment Analysis
- Text Summarization

Conclusion:

Learning about the Llama model gave us insight into how cutting-edge AI works behind the scenes. It showed us that efficient AI doesn’t always mean massive resources — smart design and transparency matter too. It was a great session to connect what we see in chatbots and smart assistants to the technology that powers them.